

Assignment 3 :

Q Trends and cloud services overview

1. Advantages of Serverless architecture -

- i) Removes server management and provisioning needs.
- ii) Automatically scales to demand, which enhances efficiency.
- iii) Provides a pay-as-you-go pricing model that saves costs associated with variable workloads.
- iv) Makes development cycles shorter and enables developers to concentrate on code rather than infrastructure.

2. Progressive web apps (PWAs) -

- i) Web apps that behave similarly to native applications, but through a browser.
- ii) Provide offline usage, instant load times and push notifications.
- iii) It can be installed directly on a user's device without the need to pass through app stores.
- iv) It is built with modern web technologies like service workers and manifests.

3. Role of AI and machine learning in software architecture

- i) It allows systems to analyze data, identify patterns and make intelligent decisions.
- ii) Features like predictive analytics, personalization and fraud detection rely on this.
- iii) Integration with scalable cloud platforms for data handling and model training is needed.

- iv) This makes automation more effective and user experience better with smarter applications.

Cloud Computing Service models and Applications.

- 1) Software as a service (SaaS) - Provides software via the internet, ~~users~~ ~~to~~ The applications are accessed with a web browser without any regard for installation, upkeep or infrastructure.

Uses

- i) Email services (eg- gmail, outlook)
- ii) Customer Relationship management (CRM) software (eg- salesforce)
- iii) Office software (eg- google workspace)

- 2) Platform as a service (PaaS) - provides an environment and platform upon which programmers can write, test and run programs. It includes the tools, libraries, and infrastructure needed to write but conceals the hardware administration.

Uses

- i) Developing and hosting web or mobile applications.
- ii) Automating software development and deployment.
- iii) Database and API integration.
eg- google app engine, microsoft azure app services etc.

- 3) Infrastructure as a Service (IaaS) - delivers virtualized computing resources like servers, storage and networking over the internet. The users get OS, application and configuration control.

Uses .

- i) Hosting websites and applications.
- ii) Storing and backing up data.
- iii) Configuring virtual machines for testing and development.

Eg - Amazon, microsoft azure VMs etc.